

AgriCast: AI-Powered Pest Forecasting for Resilient Maize Farming

a. Title

AgriCast: AI-Powered Pest Forecasting for Resilient Maize Farming

b. Problem Statement

Agriculture in Kenya is heavily reliant on smallholder farmers, who produce up to 75% of the country's food. However, one of the biggest threats to food security is the unpredictable surge in pest infestations, exacerbated by climate change. Invasive pests such as locusts, fall armyworms, and stem borers cause devastating losses to staple crops like maize — often without warning.

The challenge is further compounded by the lack of localized, data-driven early warning systems. Traditional approaches to pest control are reactive, expensive, and inaccessible to many farmers. Without timely interventions, outbreaks lead to reduced yields, increased poverty, and food insecurity — especially among rural communities.

c. Objectives

1. To develop an AI-powered system that predicts pest infestation risks based on real-time environmental and weather data.
2. To empower smallholder farmers with localized, actionable insights through an interactive digital platform.
3. To reduce crop losses and improve food security by providing early warning alerts for pest outbreaks.
4. To promote climate resilience through smarter, data-driven agricultural planning.

d. Methodology

- Data Collection: Integrate weather data (temperature, humidity, rainfall, wind), NDVI vegetation indices, and simulated pest occurrence datasets. Use OpenWeather API and publicly available climate sources.
- Model Development: Train machine learning models (Random Forest / XGBoost) to predict severity of key maize pests: locusts, fall armyworm, stem borer, corn earworm, and ear rot.
- Platform Implementation: Deploy the AI model using Streamlit, creating a user-friendly web app where farmers or extension officers can input weather conditions or fetch real-time city data.
- Output Delivery: The platform returns severity levels for each pest (None, Low, Medium,

High) with tailored mitigation advice, and optionally visualizes risk on a map.

- Future Enhancements: Integration with mobile devices, alerts via SMS/WhatsApp, and feedback loops to refine model accuracy.



Maize Pest and Disease Invasion Predictor

Use weather data to predict pest invasion levels and help farmers make informed pest control decisions.

Select weather input method

☐ Manual Entry

☒ Enter City

Enter city name

Kampala

Weather data for Kampala

```
{
  "city": "Kampala",
  "lat": 0.3163,
  "lon": 32.5822,
  "temp_max": 26.53,
  "temp_min": 26.53,
  "rainfall": 0,
  "humidity": 55,
  "wind_speed": 2.79
}
```

Prediction Results

 Fall Armyworm: Medium

 Ear Rot: High

 Stem Borer: None

 Corn Earworm: Low

 Locust: High

High Risk Detected

Farmers should:

- Apply preventive treatment
- Scout regularly
- Use traps and monitor frequently

Share

 Soil Moisture (%)

25.00

- +

 NDVI (0.1-0.9)

0.50

- +

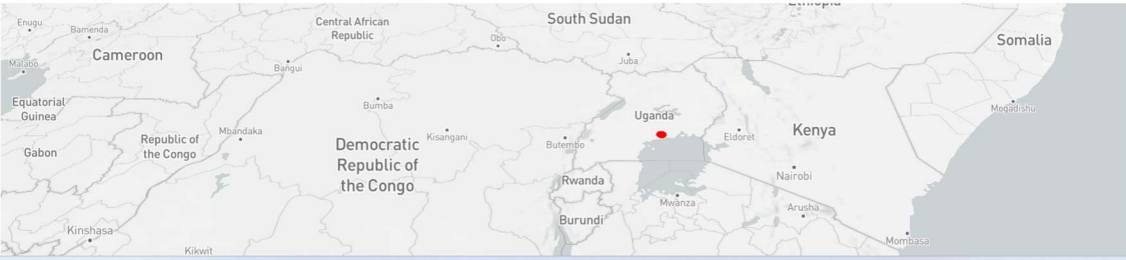
 Altitude (m)

1500.00

- +

 Predict

Maize Pest Risk Map



Team Name: AgriCast

Institution: Dedan Kimathi University of Technology

Members: Zipporah Mutua, Edward, Lawrence

Mentor: Dr. John Wandeto

Thematic Area: Climate Change and Food Security